Parity and invariants

High school · reasoning · Lesson and method · 01

Prerequisites. Even and odd integers; proof by contradiction.

Key point · Ideas and method. An invariant is a quantity or property preserved by every allowed move. If the target has a different invariant value from the initial state, it cannot be reached.
To study parity, replace integers with their remainders 0 or 1 modulo 2. Two odd integers have an even sum. Adding an even integer preserves parity.
Method: specify one move, find what it preserves, and compute the invariant at the start and at the target. An obstruction proves impossibility; the absence of an obstruction does not prove possibility.

Example 1. Start at 0 and add or subtract 6 at each move. Every reachable number is even, so 15 is unreachable. The number 14 is also unreachable, but parity alone cannot prove this: every reachable number is a multiple of 6.

Activity 1 · Explore. Write several sequences of moves, then classify the reached numbers. Find a common property more precise than parity.